

Principal Component Analysis (PCA)

By: Kinza Afzal

1 Principal Component Analysis (PCA)

1.1 Introduction

Principal Component Analysis (PCA) is a widely used **dimensionality reduction** technique in machine learning and statistics. The main idea is to reduce the number of features in a dataset while preserving as much of the data's variation as possible. PCA achieves this by identifying **principal components**, which are orthogonal directions capturing maximum variance in the data.

PCA is useful for:

- Reducing computational complexity.
- Visualizing high-dimensional data.
- Removing noise and redundant features.
- Preparing data for other machine learning algorithms.

1.2 Steps for Dimensionality Reduction using PCA

The steps for applying PCA are as follows:

1. **Standardize the data** to ensure all features contribute equally.
2. **Compute the covariance matrix** of the standardized data.
3. **Calculate eigenvalues and eigenvectors** of the covariance matrix.
4. **Sort eigenvalues in descending order** and select corresponding eigenvectors.
5. **Compute principal components** by projecting the data onto the selected eigenvectors.
6. **Reduce dimensions** by choosing the top k principal components that explain most of the variance.

1.3 Standardization

Standardization ensures that features with different scales do not dominate PCA. Common methods include:

- **Z-score standardization:**

$$z = \frac{x - \mu}{\sigma}$$

where μ is the mean and σ is the standard deviation.

- **Min-max scaling:**

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

Scales features to the $[0, 1]$ range.

- **Mean centering:**

$$x' = x - \mu$$

Centers the data around zero mean.

1.4 Variance and Covariance Formulas

- **Variance:** Measures the spread of a feature.

$$\text{Var}(X) = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2$$

- **Covariance:** Measures how two features vary together.

$$\text{Cov}(X, Y) = \frac{1}{n} \sum_{i=1}^n (x_i - \mu_X)(y_i - \mu_Y)$$

1.5 Numerical Example of PCA

Consider the following dataset with 3 samples and 2 features:

$$X = \begin{bmatrix} 2 & 0 \\ 0 & 1 \\ 3 & 1 \end{bmatrix}$$

1.5.1 Step 1: Standardize the data (Z-score)

Compute mean and standard deviation for each feature:

$$\mu_1 = \frac{2+0+3}{3} = 1.667, \quad \sigma_1 = \sqrt{\frac{(2-1.667)^2 + (0-1.667)^2 + (3-1.667)^2}{3}} \approx 1.247$$

$$\mu_2 = \frac{0+1+1}{3} = 0.667, \quad \sigma_2 = \sqrt{\frac{(0-0.667)^2 + (1-0.667)^2 + (1-0.667)^2}{3}} \approx 0.471$$

Standardized data:

$$Z = \begin{bmatrix} \frac{2-1.667}{1.247} & \frac{0-0.667}{0.471} \\ \frac{0-1.667}{1.247} & \frac{1-0.667}{0.471} \\ \frac{3-1.667}{1.247} & \frac{1-0.667}{0.471} \end{bmatrix} \approx \begin{bmatrix} 0.267 & -1.414 \\ -1.336 & 0.707 \\ 1.069 & 0.707 \end{bmatrix}$$

1.5.2 Step 2: Compute Covariance Matrix

$$C = \frac{1}{n-1} Z^T Z$$

$$C \approx \begin{bmatrix} 2.333 & 0.666 \\ 0.666 & 1.667 \end{bmatrix}$$

1.5.3 Step 3: Compute Eigenvalues and Eigenvectors

Solve $\det(C - \lambda I) = 0$ to find eigenvalues:

$$\lambda_1 \approx 2.882, \quad \lambda_2 \approx 1.118$$

Eigenvectors corresponding to each eigenvalue:

$$v_1 \approx \begin{bmatrix} 0.881 \\ 0.472 \end{bmatrix}, \quad v_2 \approx \begin{bmatrix} -0.472 \\ 0.881 \end{bmatrix}$$

1.5.4 Step 4: Compute Principal Components

Project standardized data onto eigenvectors:

$$PC = Z \cdot V$$

$$PC \approx \begin{bmatrix} -1.040 & -0.068 \\ -0.319 & 0.884 \\ 1.358 & -0.816 \end{bmatrix}$$

1.5.5 Step 5: Reduce Dimensions

To reduce to 1 dimension, choose the principal component with the largest eigenvalue (λ_1). Reduced data:

$$PC_{reduced} \approx \begin{bmatrix} -1.040 \\ -0.319 \\ 1.358 \end{bmatrix}$$

1.6 Conclusion

- PCA reduces data from 2D to 1D while retaining maximum variance. - Standardization is essential to ensure features with larger scales do not dominate PCA. - Z-score is the most commonly used standardization for PCA, though Min-Max scaling or mean-centering can also be applied depending on the dataset.

2 Detailed PCA Example with Full Computations

2.1 Dataset

Consider the dataset with 10 samples and 2 features:

Sample	x_1	x_2
1	2.5	2.4
2	0.5	0.7
3	2.2	0.9
4	1.9	2.2
5	3.1	2.3
6	2.3	3.0
7	2.0	1.6
8	1.0	1.1
9	1.5	1.6
10	1.1	0.9

Table 1: Original Data

Means and standard deviations of features:

$$\bar{x}_1 = 1.81, \quad \bar{x}_2 = 1.91$$

$$\sigma_1 = 0.78, \quad \sigma_2 = 0.84$$

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2.2 Step 1: Standardize the Data (Z-score)

Formula:

$$z_{ij} = \frac{x_{ij} - \bar{x}_j}{\sigma_j}$$

Sample	z_1	z_2
1	$\frac{2.5-1.81}{0.78} \approx 0.897$	$\frac{2.4-1.91}{0.84} \approx 0.583$
2	$\frac{0.5-1.81}{0.78} \approx -1.667$	$\frac{0.7-1.91}{0.84} \approx -1.452$
3	$\frac{2.2-1.81}{0.78} \approx 0.513$	$\frac{0.9-1.91}{0.84} \approx -1.202$
4	$\frac{1.9-1.81}{0.78} \approx 0.115$	$\frac{2.2-1.91}{0.84} \approx 0.345$
5	$\frac{3.1-1.81}{0.78} \approx 1.667$	$\frac{2.3-1.91}{0.84} \approx 0.452$
6	$\frac{2.3-1.81}{0.78} \approx 0.641$	$\frac{3.0-1.91}{0.84} \approx 1.310$
7	$\frac{2.0-1.81}{0.78} \approx 0.244$	$\frac{1.6-1.91}{0.84} \approx -0.369$
8	$\frac{1.0-1.81}{0.78} \approx -1.038$	$\frac{1.1-1.91}{0.84} \approx -0.976$
9	$\frac{1.5-1.81}{0.78} \approx -0.398$	$\frac{1.6-1.91}{0.84} \approx -0.369$
10	$\frac{1.1-1.81}{0.78} \approx -0.897$	$\frac{0.9-1.91}{0.84} \approx -1.202$

Table 2: Standardized Data

2.3 Step 2: Compute Covariance Matrix

Covariance formula:

$$\text{Cov}(X, Y) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

Standardized data matrix Z (10×2):

$$Z = \begin{bmatrix} 0.897 & 0.583 \\ -1.667 & -1.452 \\ 0.513 & -1.202 \\ 0.115 & 0.345 \\ 1.667 & 0.452 \\ 0.641 & 1.310 \\ 0.244 & -0.369 \\ -1.038 & -0.976 \\ -0.398 & -0.369 \\ -0.897 & -1.202 \end{bmatrix}$$

Covariance matrix:

$$C = \frac{1}{n-1} Z^T Z$$

Step 2a: Compute Z^T :

$$Z^T = \begin{bmatrix} 0.897 & -1.667 & 0.513 & 0.115 & 1.667 & 0.641 & 0.244 & -1.038 & -0.398 & -0.897 \\ 0.583 & -1.452 & -1.202 & 0.345 & 0.452 & 1.310 & -0.369 & -0.976 & -0.369 & -1.202 \end{bmatrix}$$

Step 2b: Multiply $Z^T Z$ ($2 \times 10 \times 10 \times 2 = 2 \times 2$):

$$Z^T Z = \begin{bmatrix} \sum z_1^2 & \sum z_1 z_2 \\ \sum z_1 z_2 & \sum z_2^2 \end{bmatrix} = \begin{bmatrix} 18.0 & 16.2 \\ 16.2 & 17.0 \end{bmatrix}$$

Divide by $n - 1 = 9$:

$$C = \frac{1}{9} \begin{bmatrix} 18.0 & 16.2 \\ 16.2 & 17.0 \end{bmatrix} \approx \begin{bmatrix} 2.0 & 1.8 \\ 1.8 & 1.889 \end{bmatrix}$$

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2.4 Step 3: Eigenvalues and Eigenvectors

Solve $\det(C - \lambda I) = 0$:

$$\begin{vmatrix} 2 - \lambda & 1.8 \\ 1.8 & 1.889 - \lambda \end{vmatrix} = 0$$

$$(2 - \lambda)(1.889 - \lambda) - 1.8^2 = 0$$

$$\lambda^2 - 3.889\lambda + (2 * 1.889 - 3.24) = 0$$

$$\lambda^2 - 3.889\lambda + 0.538 = 0$$

Solve quadratic:

$$\lambda_1 \approx 3.807, \quad \lambda_2 \approx 0.082$$

Eigenvectors (solve $(C - \lambda I)v = 0$):

$$v_1 \approx \begin{bmatrix} 0.735 \\ 0.678 \end{bmatrix}, \quad v_2 \approx \begin{bmatrix} -0.678 \\ 0.735 \end{bmatrix}$$

2.5 Step 4: Compute Principal Components

$$PC = Z \cdot V$$

Example first row:

$$PC_1 = 0.897 * 0.735 + 0.583 * 0.678 \approx 1.040$$

$$PC_2 = 0.897 * (-0.678) + 0.583 * 0.735 \approx -0.089$$

Repeat for all 10 samples:

Sample	PC1	PC2
1	1.040	-0.089
2	-2.318	0.080
3	-0.512	-0.905
4	0.300	0.115
5	2.062	-0.159
6	1.702	0.178
7	0.059	-0.381
8	-1.686	0.228
9	-0.537	-0.022
10	-0.111	-0.044

Table 3: Principal Components

2.6 Step 5: Dimensionality Reduction

- Keep only first principal component (largest eigenvalue λ_1):

$$PC_{reduced} = \begin{bmatrix} 1.040 \\ -2.318 \\ -0.512 \\ 0.300 \\ 2.062 \\ 1.702 \\ 0.059 \\ -1.686 \\ -0.537 \\ -0.111 \end{bmatrix}$$

This captures most of the variance in a single dimension.

2.7 Conclusion

- Every step is shown: standardization, covariance, Z^T , multiplication, eigenvalues, eigenvectors, projection. - PC1 captures majority of the variance; PC2 is minor. - Standardization is essential before PCA to avoid feature scale bias.

3 PCA Worked Example: 3 Samples, 3 Features

Consider the following dataset with 3 samples and 3 features:

Sample	x_1	x_2	x_3
1	2	0	1
2	0	1	2
3	1	3	1

Table 4: Original Data

4 Step 1: Compute Means

$$\bar{x}_1 = \frac{2+0+1}{3} = 1, \quad \bar{x}_2 = \frac{0+1+3}{3} = 1.333, \quad \bar{x}_3 = \frac{1+2+1}{3} = 1.333$$

5 Step 2: Deviations from Mean

Sample	$x_1 - \bar{x}_1$	$x_2 - \bar{x}_2$	$x_3 - \bar{x}_3$
1	1	-1.333	-0.333
2	-1	-0.333	0.667
3	0	1.667	-0.333

Table 5: Deviations from Mean

6 Step 3: Compute Covariances (Formula)

$$\text{Cov}(X, Y) = \frac{\sum_{i=1}^n (x_i - \bar{X})(y_i - \bar{Y})}{n - 1}$$

Step 3a: Variances

$$\text{Var}(x_1) = \frac{1^2 + (-1)^2 + 0^2}{2} = \frac{2}{2} = 1$$

$$\text{Var}(x_2) = \frac{(-1.333)^2 + (-0.333)^2 + (1.667)^2}{2} = \frac{1.777 + 0.111 + 2.778}{2} = \frac{4.666}{2} \approx 2.333$$

$$\text{Var}(x_3) = \frac{(-0.333)^2 + 0.667^2 + (-0.333)^2}{2} = \frac{0.111 + 0.445 + 0.111}{2} = \frac{0.667}{2} \approx 0.333$$

Step 3b: Covariances

$$\text{Cov}(x_1, x_2) = \frac{1 \cdot (-1.333) + (-1) \cdot (-0.333) + 0 \cdot 1.667}{2} = \frac{-1.333 + 0.333 + 0}{2} = -0.5$$

$$\text{Cov}(x_1, x_3) = \frac{1 \cdot (-0.333) + (-1) \cdot 0.667 + 0 \cdot (-0.333)}{2} = \frac{-0.333 - 0.667 + 0}{2} = -0.5$$

$$\text{Cov}(x_2, x_3) = \frac{(-1.333)(-0.333) + (-0.333)(0.667) + (1.667)(-0.333)}{2} = \frac{0.444 - 0.222 - 0.555}{2} = -0.1665$$

7 Step 4: Standardize the Data (Z-score)

Z-score formula:

$$z_{ij} = \frac{x_{ij} - \bar{x}_j}{\sigma_j}, \quad \text{where } \sigma_j = \sqrt{\text{Var}(x_j)}$$

Compute standard deviations:

$$\sigma_1 = \sqrt{1} = 1, \quad \sigma_2 = \sqrt{2.333} \approx 1.528, \quad \sigma_3 = \sqrt{0.333} \approx 0.577$$

Z-score table:

Sample	$z_1 = (x_1 - \bar{x}_1)/\sigma_1$	$z_2 = (x_2 - \bar{x}_2)/\sigma_2$	$z_3 = (x_3 - \bar{x}_3)/\sigma_3$
1	$1/1 = 1.000$	$-1.333/1.528 \approx -0.873$	$-0.333/0.577 \approx -0.577$
2	$-1/1 = -1.000$	$-0.333/1.528 \approx -0.218$	$0.667/0.577 \approx 1.155$
3	$0/1 = 0.000$	$1.667/1.528 \approx 1.091$	$-0.333/0.577 \approx -0.577$

Table 6: Standardized Data (Z-score) for all samples and features

The standardized matrix Z is:

$$Z = \begin{bmatrix} 1.000 & -0.873 & -0.577 \\ -1.000 & -0.218 & 1.155 \\ 0.000 & 1.091 & -0.577 \end{bmatrix}$$

This matrix will be used for PCA transformation and computation of principal components.

8 Step 5: Covariance Matrix

$$C = \begin{bmatrix} \text{Var}(x_1) & \text{Cov}(x_1, x_2) & \text{Cov}(x_1, x_3) \\ \text{Cov}(x_2, x_1) & \text{Var}(x_2) & \text{Cov}(x_2, x_3) \\ \text{Cov}(x_3, x_1) & \text{Cov}(x_3, x_2) & \text{Var}(x_3) \end{bmatrix} = \begin{bmatrix} 1 & -0.5 & -0.5 \\ -0.5 & 2.333 & -0.167 \\ -0.5 & -0.167 & 0.333 \end{bmatrix}$$

9 Step 6: Determinant of Covariance Matrix

$$\det(C) = a(ei - fh) - b(di - fg) + c(dh - eg)$$

$$\begin{aligned}\det(C) &= 1(2.333 \cdot 0.333 - (-0.167) \cdot (-0.167)) \\ &\quad - (-0.5)((-0.5) \cdot 0.333 - (-0.167) \cdot (-0.5)) \\ &\quad + (-0.5)((-0.5) \cdot (-0.167) - 2.333 \cdot (-0.5))\end{aligned}$$

Step by step:

$$1(0.7769 - 0.0279) - (-0.5)(-0.1665 - 0.0835) + (-0.5)(0.0835 + 1.1665) = 0.749 - 0.125 - 0.625 \approx -0.001$$

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10 Step 7: Eigenvalues and Eigenvectors

Solve:

$$\det(C - \lambda I) = 0$$

$$\begin{vmatrix} 1 - \lambda & -0.5 & -0.5 \\ -0.5 & 2.333 - \lambda & -0.167 \\ -0.5 & -0.167 & 0.333 - \lambda \end{vmatrix} = 0$$

Cubic equation (expanded):

$$\begin{aligned}0 &= (1 - \lambda)((2.333 - \lambda)(0.333 - \lambda) - (-0.167)(-0.167)) \\ &\quad + 0.5(-0.5(0.333 - \lambda) - (-0.167)(-0.5)) \\ &\quad - 0.5((-0.5)(-0.167) - (2.333 - \lambda)(-0.5))\end{aligned}$$

Numerical solution:

$$\lambda_1 \approx 2.54, \quad \lambda_2 \approx 0.93, \quad \lambda_3 \approx 0.19$$

Corresponding normalized eigenvectors:

$$v_1 \approx \begin{bmatrix} -0.51 \\ 0.85 \\ -0.12 \end{bmatrix}, \quad v_2 \approx \begin{bmatrix} 0.69 \\ 0.50 \\ -0.52 \end{bmatrix}, \quad v_3 \approx \begin{bmatrix} 0.51 \\ 0.15 \\ 0.84 \end{bmatrix}$$

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11 Step 8: Feature Vector Construction

$$V = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix} = \begin{bmatrix} -0.51 & 0.69 & 0.51 \\ 0.85 & 0.50 & 0.15 \\ -0.12 & -0.52 & 0.84 \end{bmatrix}$$

12 Step 9: Transform Data (Principal Components)

Multiply each row of Z by each eigenvector:

$$Z_{reduced} = Z \cdot V$$

Step by step (row 1):

$$PC1_1 = 1 \cdot (-0.51) + (-0.872) \cdot 0.85 + (-0.577) \cdot (-0.12) \approx -0.51 - 0.741 + 0.069 = -1.182$$

$$PC2_1 = 1 \cdot 0.69 + (-0.872) \cdot 0.50 + (-0.577) \cdot (-0.52) \approx 0.69 - 0.436 + 0.300 = 0.554$$

$$PC3_1 = 1 \cdot 0.51 + (-0.872) \cdot 0.15 + (-0.577) \cdot 0.84 \approx 0.51 - 0.131 - 0.485 = -0.106$$

Similarly compute for other rows:

$$Z_{reduced} \approx \begin{bmatrix} -1.182 & 0.554 & -0.106 \\ 0.662 & -0.926 & 0.528 \\ 0.520 & 0.372 & -0.422 \end{bmatrix}$$

13 Step 10: Explained Variance

$$\sum \lambda_i = 2.54 + 0.93 + 0.19 = 3.66$$

$$PC1: \frac{2.54}{3.66} \times 100 \approx 69.4\%$$

$$\text{PC2: } \frac{0.93}{3.66} \times 100 \approx 25.4\%$$

$$\text{PC3: } \frac{0.19}{3.66} \times 100 \approx 5.2\%$$

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14 Conclusion

- Step-by-step PCA with all decimals shown.
- Covariance matrix calculated using formula.
- Eigenvalues and eigenvectors computed.
- Transformation shows principal component scores.
- Explained variance shows how much each PC contributes.